From Mobility to Accessibility - Examining How European Urban Policies Balance Connectivity and Climate Neutrality: The Case of Milan

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1. Introduction

Mobility has been a central aspect of urban development from the beginning. People, goods, services and information move to grow the cities; transport networks enable access to employment, education, healthcare and leisure activity. The mobility of people has also been an important factor in enhancing the economic integration and territorial cohesion process at the European level, linking regions and overcoming locational marginalization. Over several decades, the goal of improving mobility was considered largely positive as it increased the competitiveness of the economy and enhanced citizens' opportunities.

But the connection between mobility and urban development is becoming increasingly complicated in the era of climate change. Mobility is crucial for economic productivity and social inclusion, but *transport is also one of the major sources of greenhouse gas emissions in the European Union.*

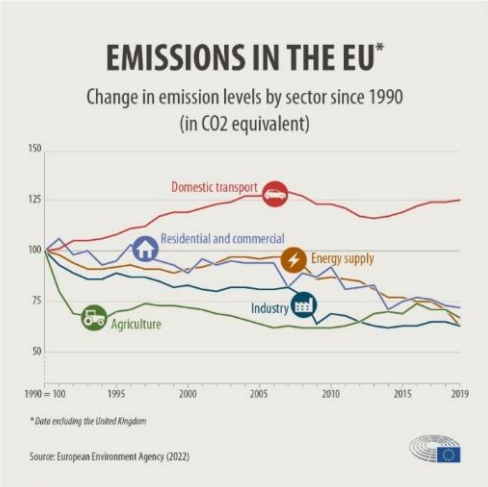


Fig. 1. Evolution of CO_2 Emissions in the EU by Sector

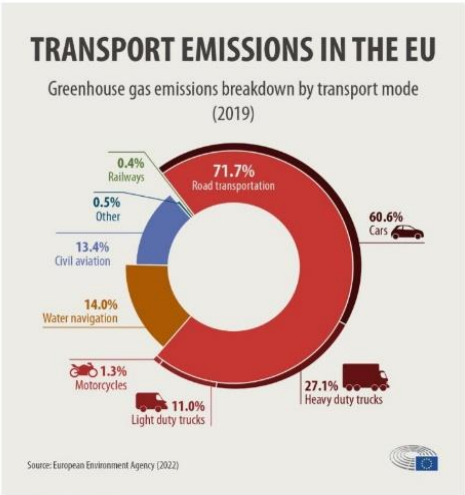


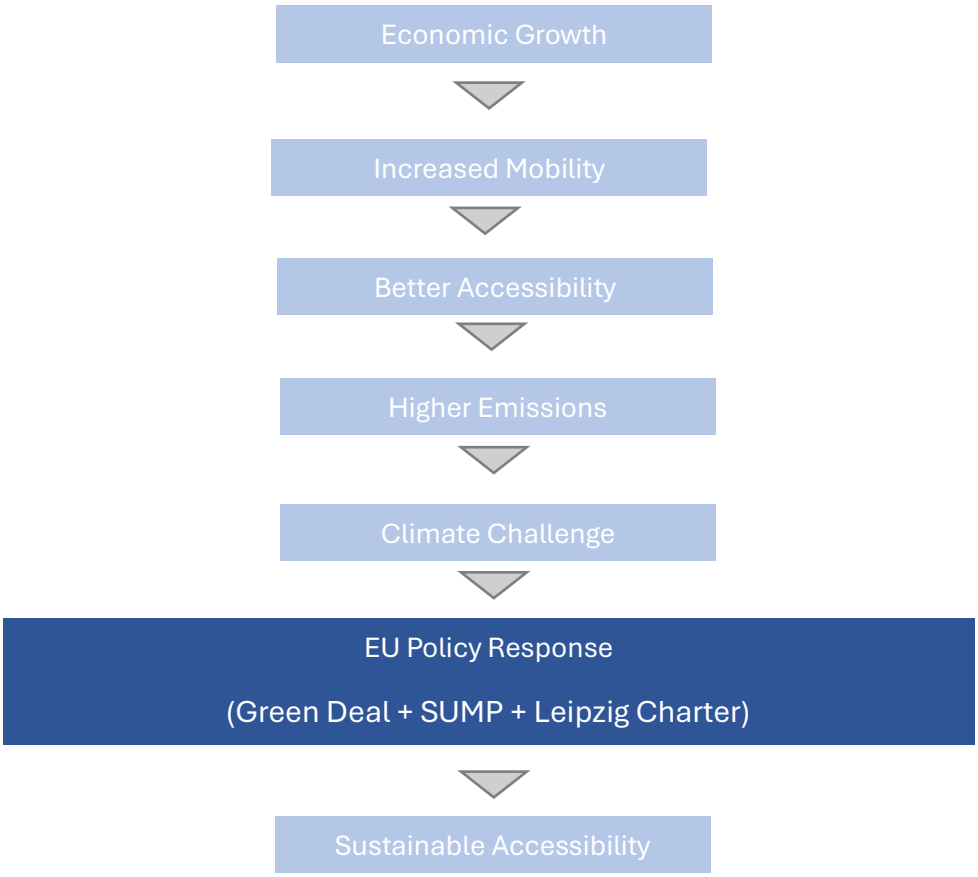
Fig. 2. Transport emissions in the EU

According to figure 1 & 2, ***transport is responsible for about 25% of EU GHGs, and road transport of about 72% of transport GHGs***. Road transport continues to generate significant levels of carbon emissions, air pollution and energy consumption. The challenge for European cities is to reconcile access and connectivity with the need to minimise the environmental footprint of mobility in the context of their ambitious climate goals, such as achieving ***climate neutrality by 2050.***

This challenge is more significant in the context of the urban policy of the European Union. Promoting territorial cohesion, accessibility and regional connectivity are still in the agenda of European policies on one hand. Citizens must be able to access jobs, services and opportunities regardless of where they live. However, the European Green Deal and policies on sustainability demand a considerable cut in carbon emissions, particularly from transportation. It is increasingly necessary to solve urban planning problems that have to simultaneously solve both of these objectives rather than considering them as two policy areas.

In this context, the city of Milan (Italy) is taken as a case study to illustrate how a European city is trying to reconcile the rising demands for mobility with the mitigation of climate change. While Milan has concentrated on improving the sustainability of existing mobility systems, when it is compared to Paris which has gained international recognition for its accessibility-oriented approach through the 15-minute city model. The paper aims to look at the experiences of Milan and Paris in the context of the European Green Deal, Sustainable Urban Mobility Plans (SUMPs), and the New Leipzig Charter and how it can guide more sustainable and inclusive mobility planning in India.

The analysis is part of a larger discussion that exists in urban planning about the relationship of mobility, accessibility and sustainability. Instead of asking whether mobility should be decreased, the paper asks the question of how European cities can keep access to opportunities while minimizing the environmental impacts of mobility.



2. Policy Framework

The EU has put in place an integrated policy framework between mobility demand and climate change with the goal of reducing the negative impacts of transport on climate change. To promote low-carbon, accessible and resilient mobility systems, cities are encouraged to build them through integrated planning and governance, with key policy initiatives under the European Union's European Green Deal, Sustainable Urban Mobility Plans (SUMPs) and the New Leipzig Charter.

2.1 The European Green Deal

The European Green Deal (EGD) adopted by the European Commission in 2019 is the key EU policy to reach climate neutrality by 2050 (European Commission, 2019). The Green Deal is not just an environment policy, but a package of measures for the European economy to transition towards a more sustainable development model. It covers various areas of activity such as energy, industry, agriculture, construction and transport. Transport plays a special role in the Green Deal, as one of the biggest sources of greenhouse gas emissions in Europe. The European Commission (2019) states that significant transformation of mobility of people and goods between cities and regions are needed to become climate neutral. It aims to decarbonise mobility without carbon, by favouring cleaner technologies, public transport, active mobility and integrated urban development.

2.2 Sustainable Urban Mobility Plans (SUMPs)

The Green Deal sets out long-term climate targets, whereas ***Sustainable Urban Mobility Plans (SUMPs)*** give a practical plan for sustainable mobility at the local level. SUMPs were developed as part of the limitations of traditional transport planning, which tended to concentrate on road capacity and traffic efficiency and vehicle movements. The main innovative aspect of SUMPs is the paradigm change from mobility planning to accessibility planning. The SUMPs question is not "how can traffic flow faster", but "how can people more easily reach employment, education, healthcare, public services and recreational opportunities". SUMPs is therefore to encourage integrated planning strategies, integrating transport with land use, environmental and social considerations. Key principles which are being applied include public participation, multimodal transport systems, environmental sustainability and social inclusion. The importance of walking, cycling and public transport is also emphasised as it is seen as reducing emissions, as well as enhancing urban liveability and accessibility.

2.3 The new Saxons of Leipzig

The New Leipzig Charter that was adopted in 2020 offers a wider perspective on sustainable urban development in Europe. The Charter builds on existing EU urban policy projects and highlights the transformative role of cities in addressing current challenges, including climate change, social inequality and economic restructuring (European Commission, 2020). The Charter is structured around three inter-linked principles, the Just City, the Green City and the Productive City. The New Leipzig Charter is especially relevant to this paper for its focus on integrated urban

development. Instead of tackling these three aspects of mobility, housing, environment and economic development independently, the Charter urges us to consider them as closely intertwined parts of urban systems.

The Charter is hence a valuable governance instrument to tackle the challenges of the European Green Deal and put into practice through SUMPs. Combined, these policies point towards a larger paradigm change in European urban planning from **maximisation of movement to more accessible, sustainable and inclusive cities.**

Table 1. Comparison of the EU policies and its relevance with Milan

Policy	Main Objective	Mobility Perspective	Sustainability Perspective	Relevance to Milan
European Green Deal	Climate neutrality by 2050	Maintain connectivity through cleaner transport	Reduce emissions from transport sector	Supports Area B, Area C and sustainable mobility
SUMP	Sustainable accessibility	Improve access rather than movement	Modal shift and reduced emissions	Influences mobility planning in Milan
New Leipzig Charter	Integrated urban development	Ensure accessibility and inclusion	Promote green and resilient cities	Supports integrated planning and governance

The next sections look at the implementation of these policy principles in the case of Milan and at the approaches of other European cities, in particular Paris.

3. Thematic Analysis

3.1 Milan and the Mobility–Sustainability Challenge

Milan represents one of the most significant economic and metropolitan centres in Europe. As Italy's financial capital, the city attracts hundreds of thousands of daily commuters from across Lombardy and beyond, functioning as a major hub for employment, education, commerce, and innovation (Comune di Milano, 2024). Accessibility and effective mobility systems are vital to the city's economic success and link the community to opportunities.

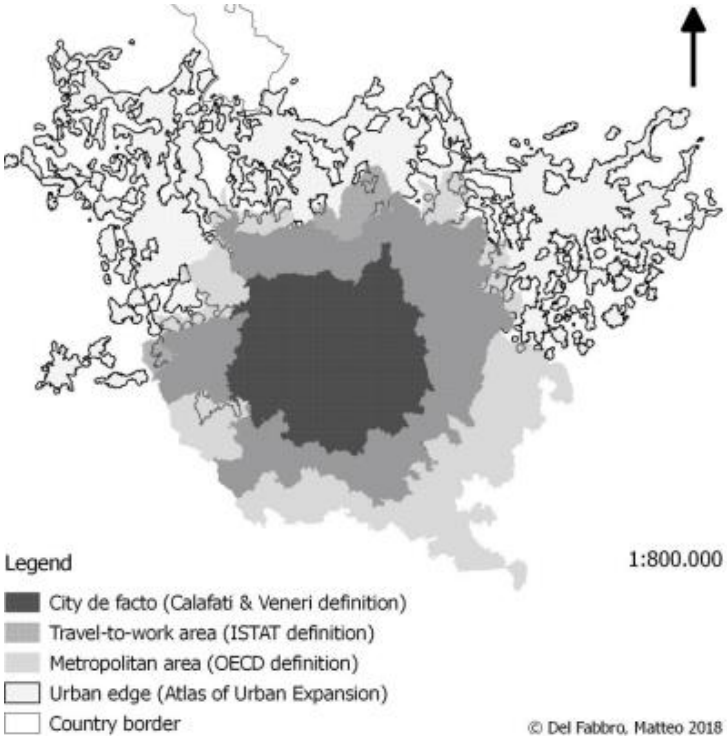


Fig. 3 . Functional territorial scales of Milan.

While mobility has helped the economies grow, the regions be more competitive, but it has also facilitated environmental degradation and carbon emission increases. Milan is therefore confronted with a key planning challenge: how to keep its level of accessibility and economic productivity high and limit the environmental costs of mobility?

Milan has tried to balance both mobility and sustainability, rather than considering them competing targets. The city is aware that the decrease of accessibility would have a negative impact on economic competitiveness and social inclusion. Rather, it aims to minimize the environmental impacts of mobility while retaining jobs, services, education and urban opportunities. This change can be viewed as a shift from **mobility-based planning to accessibility-based planning**; instead of creating the maximum possible mobility, the focus is on creating the maximum possible sustainable accessibility (Banister, 2008; Geurs & van Wee, 2004).

3.2 How Milan Translates EU Sustainability Policies into Local Action

In order to tackle these challenges, Milan has set out a number of planning strategies, all of which are in line with the overarching European sustainability goals. The city's main plan is the Piano Urbano della Mobilità Sostenibile (PUMS) – Sustainable Urban Mobility Plan. The PUMS is the city's long-term mobility plan and has been used to implement the European principles of policy into local action (Comune di Milano, 2024).

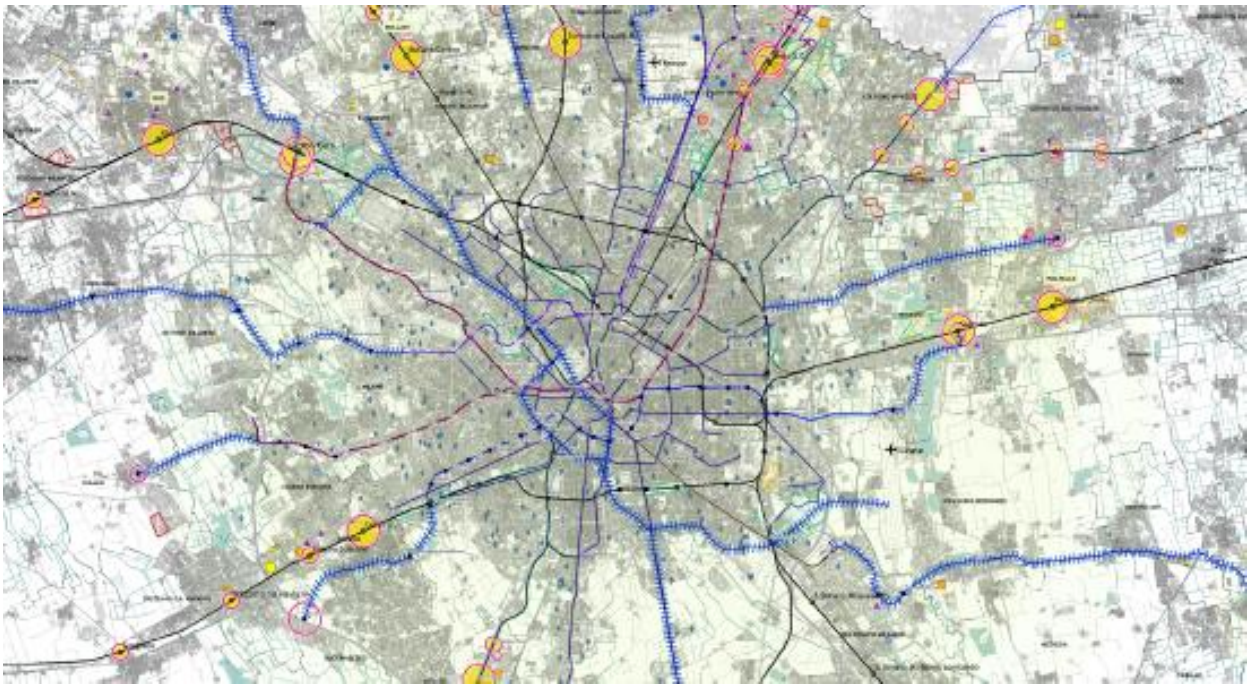


Fig.4 . Metropolitan public transport and mobility network under the Milan Metropolitan Sustainable Urban Mobility Plan (PUMS).

Three key European policies are reflected in the influence on the policies within Milan. The European Green Deal sets the overall goal of emission reduction and toward climate neutrality (European Commission, 2019). The planning methodology adopted in Sustainable Urban Mobility Plans (SUMPs) gives priority to public transport, walking, cycling and integrated mobility systems (European Commission, 2020). Compact, inclusive and environmentally sustainable urban development, based on a multi-agency governance approach and place-based planning, is encouraged by the New Leipzig Charter (2020).

3.2.1 Area C: Managing Access Rather than Restricting It



Fig 5 . Area B and Area C restricted traffic zones in Milan.

One of Milan's most significant mobility policies is Area C, a congestion charging zone introduced within the historic city centre. The objective of Area C is to decrease the number of private cars and their impact on the environment by incentivizing public transportation, walking and cycling (Comune di Milano, 2024). Notably, the policy is not limiting access per se. Access to employment, education and services is maintained but promoted via more sustainable modes of transport. Likewise, Area B is one of the largest low-emission zones in Europe, and limits access for polluting vehicles in a larger metropolitan area. Areas C and B show the implementation of the Green Deal goals at the local level, with the aim of lowering emissions and keeping areas accessible (Comune di Milano, 2024).

Investment in public transport is also a key element in Milan's strategy. The city has been able to increase metro, tram and regional rail services under PUMS, such as the recently completed M4 metro line. These investments increase access and the non-dependence on private cars. Meanwhile, the city of Milan has also been developing bicycle-friendly infrastructure and pedestrian-friendly public spaces, especially after the COVID-19 pandemic. These interventions are similar to SUMP principles, which emphasise active mobility and multi modal transport networks (European Commission, 2020).

3.2.2 Forestami and Metropolitan Sustainability

Milan's sustainability plan goes beyond its transport system. The Forestami initiative aims to plant 3 million trees throughout the metropolitan region by 2030, highlighting the city's approach to climate adaptation, environmental resilience, and biodiversity enhancement through mobility planning (Forestami, 2024). This mirrors the principle of integrated planning that is encouraged by the New Leipzig Charter, which takes an integrated approach to planning for transport, public health, environmental quality and urban resilience (New Leipzig Charter, 2020).

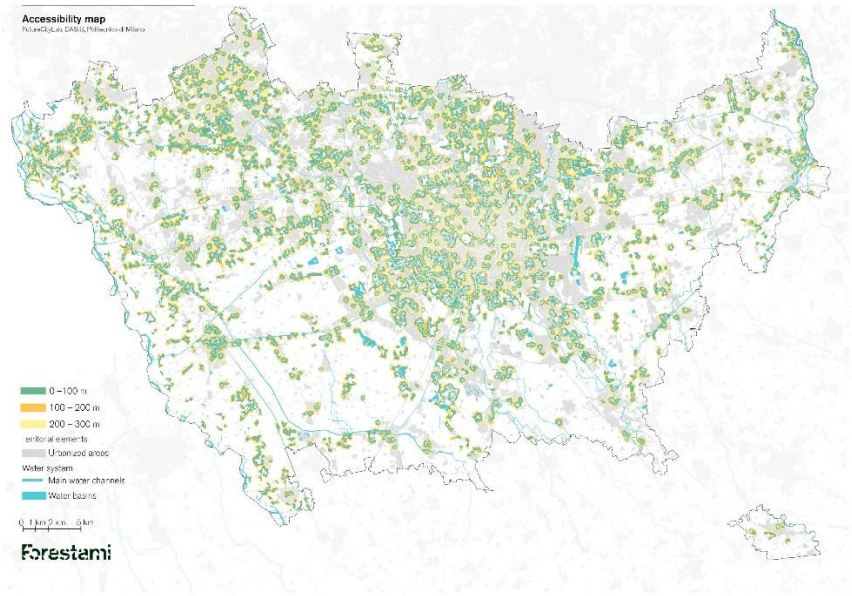


Fig 6. Forestami accessibility map (Future City Lab, Department of Architecture and Urban Studies (DAStU), Politecnico di Milano)

3.3 Governance, Policy Effectiveness and Remaining Challenges

Milan's sustainable mobility strategy is a complex governance model with a number of actors involved. The Municipality of Milan is leading the development and implementation of PUMS, Area C, cycling infrastructure projects and urban public space interventions. In addition to the local aspects covered by the Metropolitan City of Milan, the Region of Lombardy provides support for the planning of transport and infrastructure investments and communicates on the wider territorial aspects related to commuting and regional accessibility (Comune di Milano, 2024). Some evidence indicates that Milan's policies have had positive impact. In the city centre, Air quality has improved in area C, with a reduction of traffic volumes, and in area B, with a reinforcement of the measures to reduce vehicle emissions (Comune di Milano, 2024).

Increasing investments in public transport and active mobility have contributed to improving sustainable access and provided people with better access to opportunities that have lower environmental impacts. However, there are still many difficulties to overcome. The commuter flows to and from the municipality of Milan are still very high, with many originating from neighboring municipalities where other forms of transport might not be as competitive as private cars. This makes people permanently reliant on mobility which can only be solved by municipal measures. There are also concerns around social equity, as congestion charging and low emission zones may disproportionately impact low-income individuals (Banister, 2008). In addition, multi-institutional coordination can pose implementation difficulties, which are part of the broader governance issues in metropolitan regions (New Leipzig Charter, 2020).

4. Comparative Analysis: Milan and Paris

The European Green Deal, Sustainable Urban Mobility Plans (SUMPs) and the New Leipzig Charter share a common objective: improving accessibility while reducing the environmental impacts of urban mobility. The policies do not oppose economic competitiveness and the climate neutrality, but rather encourage integrated planning solutions that integrate transport, land use and environmental policies.

The cases of Milan and Paris illustrate how different European cities are translating these objectives into practice while responding to distinct urban and metropolitan challenges.

4.1 Milan: Decarbonising Mobility in a Metropolitan Region

Milan serves as a particularly relevant case-study since it contains all the tensions that are part and parcel of contemporary European policies. The city is at the centre of the metropolitan region with over three million inhabitants, and it is heavily dependent on the flows of mobility that happen daily and on the connectivity of the region (Città Metropolitana di Milano, 2024).

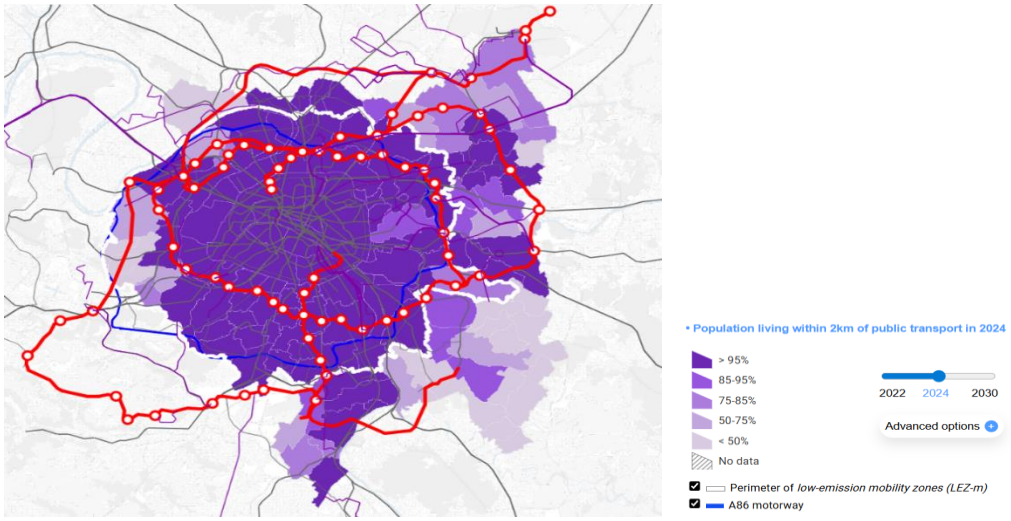


Fig 7. Public transport accessibility in the Milan metropolitan area

Accessibility is then not only a transport problem, but a condition that is essential for economic productivity and competitiveness. Milan is a large, metropolitan city that works beyond the municipal boundaries, where interventions can be implemented in the neighbourhood can help improve accessibility. There are large mobility needs into the city from outlying municipalities due to thousands of commuters travelling in each day. Given this context, it is not easy to directly reduce travel demand. Rather, Milan has prioritised the environmental aspects of mobility and aimed to decrease its effects while keeping things accessible.

4.2 Proximity: Accessibility in Paris

A new approach to planning is used to deal with the challenge in Paris. The city increasingly tries to decrease the need for mobility, rather than making it more sustainable. The idea is similar to the **15-minute city** concept which was created by Carlos Moreno and implemented in the City of Paris (Moreno et al., 2021). The 15 minute city philosophy is to connect all aspects of a city's daily life within walking and cycling zones from home to work, school, community and healthcare, retail and recreation within a 15 minute radius. Accessibility is then made by being close to, not more mobile. This is very similar to the European Green Deal objectives to reduce use of motorized transport and emissions. It also embodies the same concept of accessibility SUMPs promote, but in a different way; namely via urban form and land-use planning, rather than transport infrastructure.

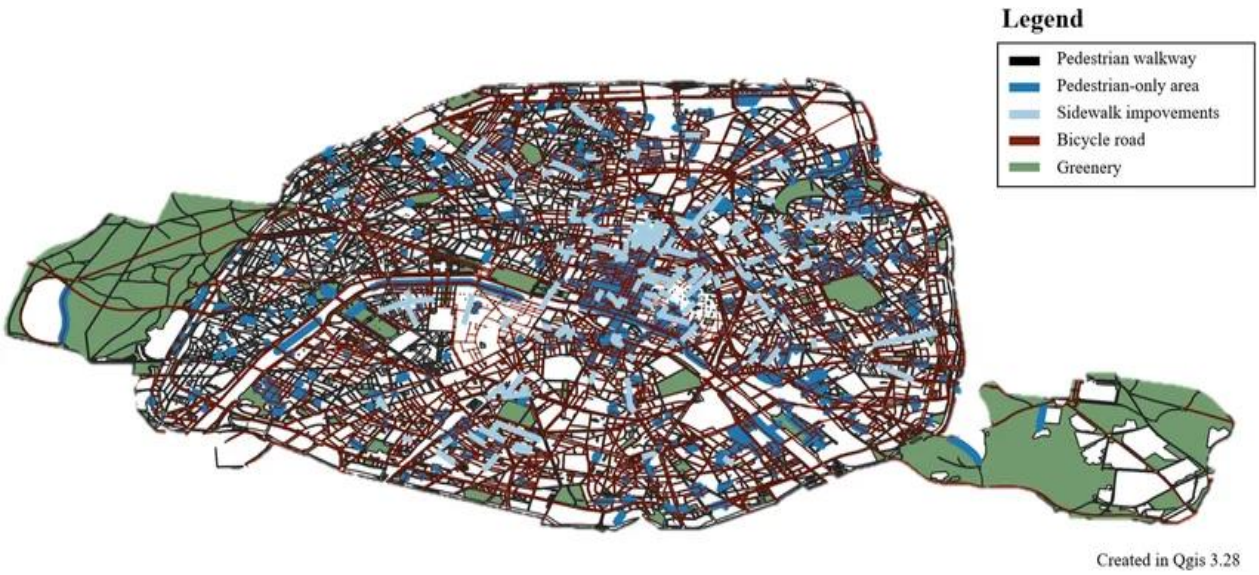


Fig 8 - Paris 15-Minute City Accessibility Map

In order to reach this goal, Paris has implemented large-scale bicycle infrastructure, pedestrianization and neighbourhood-based actions. Road space has been dedicated to cycling and walking while the local service has been enhanced to lessen travel needs. The city is also very active in bicycle infrastructure and has developed one of the densest urban bicycle networks in Europe. The New Leipzig Charter is highlighted in the actions of the city of Paris to make neighbourhoods more liveable, more inclusive and more green. The goals of the Just City and Green City are fulfilled by public space enhancements, investments in active mobility and local accessibility measures. But there are challenges facing the city as well. Affordability and social equity is a concern as property values rise in very accessible neighbourhoods. Moreover, a significant number of workers still travel from outside the city to work, and this fact testifies to the failures of place-based planning to tackle metropolitan mobility patterns.

Paris is working on implementing the New Leipzig Charter, in the form of greener, more inclusive and more liveable neighbourhoods. All measures of the public space improvements and investments in active mobility and local accessibility measures support the Just City and Green City objectives. But there are issues in the city, as well. With rising city housing values in well-connected areas, affordability and social equity are emerging as issues. Furthermore, there are still some workers, who travel to work from outside the city limits, highlighting the importance of strategies that extend beyond urban limits.

4.3 Comparing the Two Approaches

Two different but complementary approaches to the challenge of mobility and sustainability are illustrated by Milan and Paris. Milan's priorities are the implementation of the congestion charging system in the city, the expansion of the low-emission zones and investments in sustainable mobility in the metropolitan area, to reduce the environmental impacts of the mobility services offered in the city. The themes of proximity, mixed-use and neighbourhood accessibility are major issues in

the Paris travel demand reduction strategies. Both methods do not make mobility a thing of the past. Instead, both want to tweak the equation of access and mobility. Sustainable connectivity is a key element of Milan's values, while local accessibility is a key element of Paris's values.

While both are aligned with the goals of the European Green Deal, SUMPs and the New Leipzig Charter, they still struggle with the aspects of mobility and social equity as well as governance at the metropolitan level. Please insert Figure 9 (Milan–Paris Comparative Accessibility Model): The experience in Milan and Paris is thus an indication that there is no single way to sustainable urban mobility. European urban policy, instead, increasingly works towards providing opportunities of social inclusion, economic productivity and environmental sustainability through mobility. The question is not if mobility should be possible, but how accessibility will be organised within the boundaries of the environment that must be developed in the context of a climate neutral mobility.

Table 2. Milan and Paris: Two Approaches to Sustainable Accessibility

Dimension	Milan	Paris
Main Challenge	Metropolitan commuting and regional accessibility	Car dependency and urban liveability
Planning Focus	Sustainable mobility	Local accessibility
Key Strategy	Public transport, Area C, Area B	15-minute city
Scale of Intervention	Metropolitan region	Neighbourhood and city scale
Primary Objective	Reduce emissions from movement	Reduce the need for movement
Main Risk	Continued commuter dependence	Gentrification and uneven accessibility

This comparison highlights an important insight for European urban policy. Cities may pursue sustainability through different pathways depending on their spatial structure and governance context. What matters is not whether a city follows the Paris model or the Milan model, but whether it succeeds in improving accessibility while reducing environmental impacts.

4.4 Lessons for Indian Cities: The Case of Ahmedabad

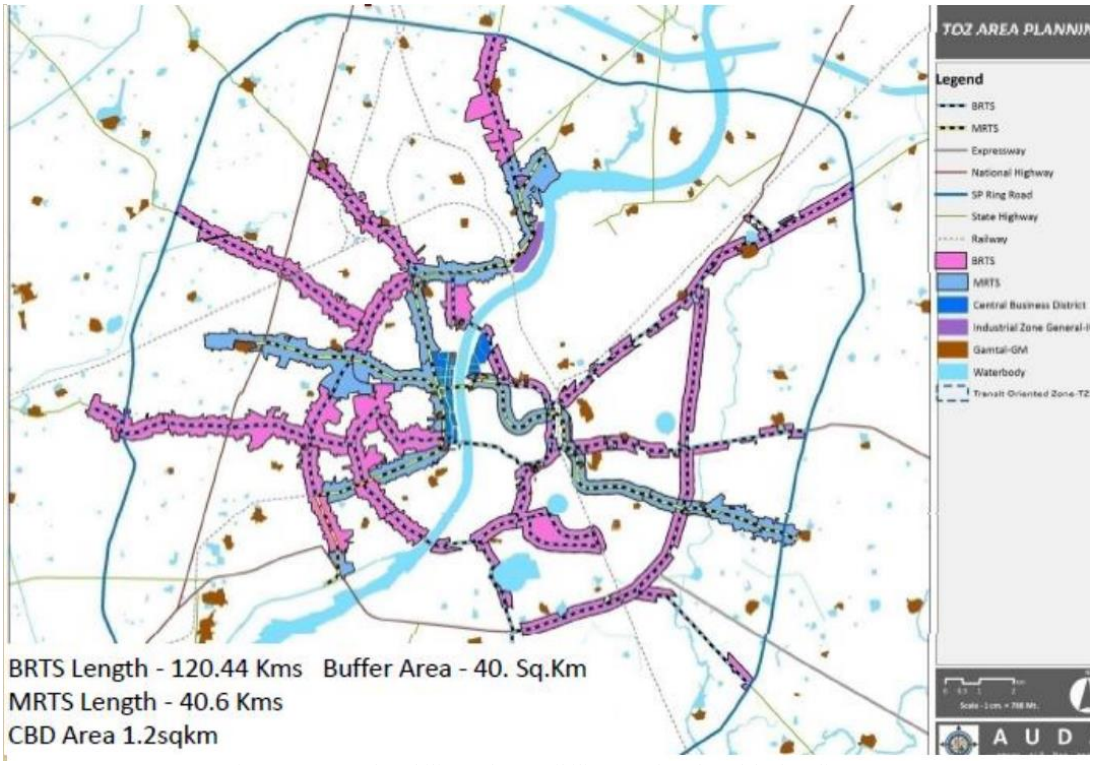


Fig 9 - Integrated Mobility and accessibility map for Ahmedabad, India

The European experience can also be instructive on the other aspects of rapid urbanisation in non-European cities. In this respect Ahmedabad, which is also a big economic centre, is interesting as comparison to Milan, the city which is facing growing mobility needs.

In the last 20 years Ahmedabad has undertaken various transport schemes such as the Bus Rapid Transit System (BRTS), development and implementation of metro rail and transit-oriented development policies. The projects have enhanced the mobility of inhabitants and facilitated access to other transport modes. But challenges like urban growth and congestion, private vehicle ownership etc. remain the strain on the city's transport system. Indian cities have relatively compact city forms and mixed land uses which are conducive to accessibility, unlike many European cities. But the benefits of urbanisation are being jeopardised by the rising population in the city and increasing reliance on cars.

First, planning for transport should be about accessibility – and not just about traffic capacity. A narrow focus on vehicle speeds can disguise more important issues of access to opportunities.

Secondly, there is a need to integrate land-use planning and transport planning. The 15-minute city has been successfully implemented in Paris showing the value of local accessibility, and regional accessibility is key in Milan.

Third, sustainability policies have to be accompanied by governance reforms. Implementation should be coordinated at multiple institutions and territorial levels.

Last, social equity must always be a key principle of sustainable mobility strategies. Jobs, education and public services should not be dependent upon private motor vehicles.

The challenge for Ahmedabad is not the same as in European cities. However, the overall question still remains the same: how to ensure access without compromising the environment?

5. Policy Recommendations

Both the Milanese and Parisian examples illustrate how European cities are moving toward an integrated approach which combines planning, climate action, land-use planning and social inclusion. Yet, comparison between the two reveals that major challenges remain. Despite a favorable policy framework provided by Europe, implementation at the city level faces political, social and spatial barriers. To better achieve balance between accessibility, economic competitiveness and climate neutrality, a number of policy adjustments could be made:

5.1 From mobility indicators to accessibility indicators

Transport planning relies on indicators such as traffic speed, flows and capacity. Although informative in explaining traffic and transport system performance, they are not indicators of access to jobs, healthcare, education, etc. The policy framework in European cities needs to place greater emphasis on accessibility indicators: metrics that gauge the efficiency of the mobility system to transport people to services and activities, whether they are jobs, education or health-care facilities.

Accessibility indicators already appear within the principles of SUMPs and need to be strongly implemented, for example with a measure on 'percentage of residents who can access employment/healthcare facilities/public transport within [certain travel time].' For Milan, this shift would take the focus off of the debate about congestion and, instead, ask whether or not parts of the metropolitan region are or are not adequately accessing opportunities.

5.2 Integrate Land Use and Mobility Planning

The contrast between Paris and Milan illustrates that policies focused solely on decreasing emissions from transport can not completely solve sustainability issues. Access is not only a function of the transport system; it is influenced by where people work, live and where services are provided. The concept of a 15 minute city has proved successful, and while the Metropolitan structure of Milan is far different than the Parisian region, it illustrates how access to daily needs is becoming increasingly important. Future city planning will need to foster opportunities to decrease the need to travel at the metropolitan scale through neighborhood based service provision, mixed-use development, and development concentrated around transport nodes. Although this

does not means elimination of cross-city transit, but an opportunity to allow people to access services at the neighborhood level as well. For fast growing metropolitan areas, integrating land-use planning with mobility planning has the potential to curb reliance on private vehicles for daily trips and to increase quality of life.

5.3 Promote Just Transitions

The case of both Milan and Paris highlight that policies designed to achieve sustainability can have detrimental impacts on certain social groups if they are not created with social inclusion in mind. Congestion charging schemes and low emission zones, for example, can disproportionately affect lower-income populations if accessible alternatives are not available. "Just City," as promoted in the New Leipzig Charter, prioritizes environmental and social justice throughout transitions to sustainability. Therefore, policy initiatives targeting transport have to include efforts to provide access and affordability to public transportation as a means to ensure the inclusion of all social groups. This could be manifested through things like reduced fare schemes for disadvantaged communities, increased investment in public transport facilities serving low-income neighborhoods, and improved accessibility for peripheral regions. Ultimately, sustainable mobility can only become a reality if it is seen to be fair for everyone.

5.4 Integrate Climate Adaptation and Mobility Planning

It becomes evident from the case of Milan how sustainable development is not merely an issue of transport but of a comprehensive citywide agenda. Policies like Forestami suggest that climate adaptation, biodiversity, public health, and environmental quality should not be treated separately from mobility and accessibility. Mobility planning needs to be integrated with climate adaptation strategies. Green infrastructure can both promote cycling and walking, decrease temperatures in urban heat islands and enhance quality of public space. Public transit investment, as well, needs to be seen as not only an investment in mobility but an investment in greater social and environmental benefits. Although this integration appears in European policies, and the Green Deal emphasizes a whole-systems approach to sustainability, it still requires greater integration at the city level.

6. Conclusion

European cities are under great pressure to achieve sustainable mobility which can also be economic and territorially coherent in one hand, and takes climate change and carbon emission reduction into account on the other. It is an undeniable factor that transport is fundamental to the life of the cities and regions; however its environmental impacts can put considerable strain on contemporary sustainability policies. This paper tried to understand how EU addresses these issues, examining three specific policy frameworks: the European Green Deal, the Sustainable Urban Mobility Plans (SUMPs) and the New Leipzig Charter, as representative of a further deepening in EU urban policy-making moving beyond physical mobility-based concept of sustainability to that of accessibility-based. The frameworks in question do not necessary target

reducing the real physical movement of people and vehicles; they plan to redesign the access based on a coordination of policies on transport, land-use planning, climate action, governance.

Through Milan's case study, we tried to understand how the city has tried to achieve a reduction in the environmental impact of its transport system without damaging its role as a center for economic development. Through the implementation of policy instruments such as Area C, Area B, and the investment on the public transport, bicycle infrastructure and Forestami project, Milan has taken significant steps towards reducing the environmental impacts associated with mobility. The comparative analysis with the Paris experience shows that urban sustainability is not a single concept and that the Paris' proximity-based approach represented by the 15- minute city is not the only strategy to realize the sustainability in cities: the comparison between Paris and Milan cases suggests that although the EU policy frameworks will lead to different kinds of urban planning as a consequence of different spatial and territorial circumstances.

It is then expected that the future of urban sustainability should take a direction towards overcoming the relationship between accessibility and mobility, by foregrounding accessibility over mobility; by addressing carbon neutrality, economic development, social justice not separately but interdependently. Taking the cases of Paris and Milan as an illustration of various strategic planning approaches aimed at such integrated planning strategy. The question that faces European cities is not if space should be created for the mobility, but rather: if and how will mobilities enable people more productively and will grant people an adequate level of access to goods and services in a context of climate change and a world of scarce carbon-prone energy resources.

REFERENCES

Banister, D. (2008). The sustainable mobility paradigm. *Transport Policy, 15*(2), 73–80.

Bertolini, L. (2012). Integrating mobility and urban development agendas: A manifesto. *Journal of Transport Geography, 24*, 487–490.

European Commission. (2019). *The European Green Deal*. Brussels: European Commission.

European Commission. (2020). *New Leipzig Charter: The transformative power of cities for the common good*. Brussels: European Commission.

European Commission. (2021). *Guidelines for Developing and Implementing a Sustainable Urban Mobility Plan (2nd Edition)*. Brussels: European Commission.

European Commission. (2020). *Sustainable and Smart Mobility Strategy*. Brussels: European Commission.

Moreno, C., Allam, Z., Chabaud, D., Gall, C., & Pratlong, F. (2021). Introducing the 15-minute city: Sustainability, resilience and place identity in future post-pandemic cities. *Smart Cities, 4*(1), 93–111.

OECD. (2024). *The 15-Minute City: A Sustainable Urban Development Model*. Paris: OECD Publishing.

Pedrazzini, L. (2026). *The EU: From spatial planning to urban policies* [Course lecture]. Politecnico di Milano.

Pedrazzini, L. (2026). *Cities and the challenge of climate change* [Course lecture]. Politecnico di Milano.

Forestami (2024) – Forestami Initiative: *A project to plant 3 million trees in the Metropolitan City of Milan by 2030.* Forestami Foundation.

Geurs, K. T., & van Wee, B. (2004). Accessibility evaluation of land-use and transport strategies: Review and research directions. Journal of Transport Geography, 12(2), 127–140. https://doi.org/10.1016/j.jtrangeo.2003.10.005

Figures and Infographics References

1. European Parliament. (2023). *CO_2 emissions from cars: facts and figures (infographics).* https://www.europarl.europa.eu/topics/en/article/20190313STO31218/co2-emissions-from-cars-facts-and-figures-infographics

2. European Parliament. (2023). *CO_2 emissions from cars: facts and figures (infographics).* https://www.europarl.europa.eu/topics/en/article/20190313STO31218/co2-emissions-from-cars-facts-and-figures-infographics

3. Del Fabbro, M. (2022). Milan metropolitan area: representation and governance [Lecture slides]. University of Trieste. https://moodle2.units.it/pluginfile.php/458238/mod_resource/content/1/SocConf_Lezione%207_Del%20Fabbro.pdf

4. Centro Studi PIM. (2020, November). *Città metropolitana di Milano: collaborazione tecnico-scientifica alla predisposizione del Piano Territoriale Metropolitano (PTM), del Piano Urbano della Mobilità Sostenibile (PUMS) e delle relative Valutazioni Ambientali Strategiche (VAS) – Fase 2 (IST_11_19).* https://www.pim.mi.it/citta-metropolitana-pums-fase-2/

5. Comune di Milano. (n.d.). *Area C*. Comune di Milano. https://www.comune.milano.it/argomenti/mobilita/area-c

6. Pastore, M. C., Parenti, C. I. M., & Patetta, C. (2025). *Accessibility within 300 meters to wellbeing green spaces in the Metropolitan City of Milan* [Map]. Forestami Research, Politecnico di Milano. Retrieved from Forestami. https://forestami.org/en/mappatura/

7. MIC-HUB. (2023). How does public transport accessibility in Milan cope with future mobility scenarios? Retrieved from https://www.mic-hub.com/project/milan-public-transport-accessibility-level-ptal

8. Paris 15-minute City map- Jeong, I., Choi, M., Kwak, J., Kim, J., & Lee, S. (2023). *A comprehensive walkability evaluation system for promoting environmental benefits*. **Scientific Reports, 13**. https://doi.org/10.1038/s41598-023-42476-3

9. Ahmedabad Urban Development Authority (AUDA). (n.d.). *Transit-oriented zone (TOZ) area planning for Ahmedabad Central Business District* [Conference presentation]. Urban Mobility India Conference. https://www.urbanmobilityindia.in/Upload/Conference/eb91c55e-1023-4bd3-b870-deb15868b440.pd